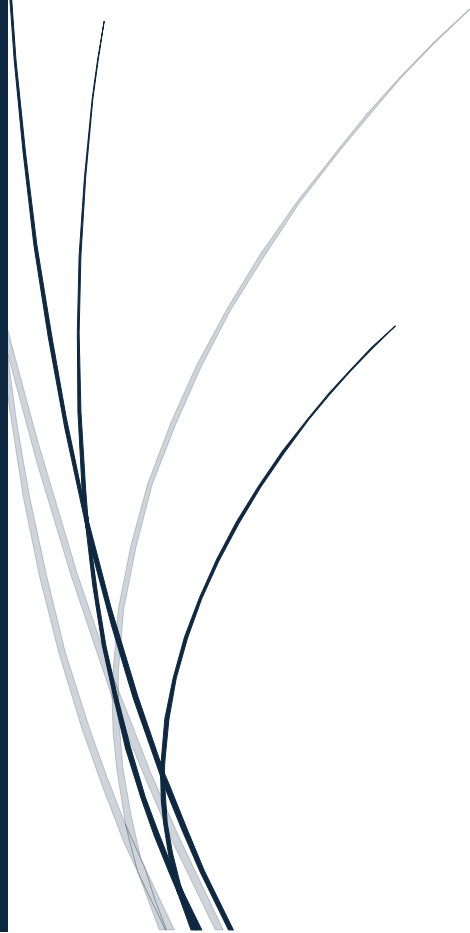




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Prescribing Trends Analysis of Antidepressant and Anxiolytic Medicines

Dataset:- Open Prescribing Dataset (NHS England)



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DATA ANALYTICS INTERN

Introduction

This project analysed prescribing trends of antidepressant and anxiolytic medicines using NHS Open Prescribing data. The objective was to identify prescribing patterns, prescribing costs, medicine utilisation trends, and regional variation across Integrated Care Boards (ICBs) and practices.

The analysis focused specifically on mental health medicines categorised under:

BNF Chapter	Category
0401	Anxiolytics / Hypnotics
0403	Antidepressants

The study included exploratory data analysis (EDA), prescribing trend analysis, forecasting, and dashboard-ready summaries.

Dataset Overview

The dataset was sourced from NHS Open Prescribing records and contained prescribing activity across England.

Dataset Characteristics

Feature	Description
Dataset Type	NHS Open Prescribing
Data Granularity	Practice-level prescribing
Geographic Coverage	NHS England
Time Variable	YEAR_MONTH
Organisational Level	ICB and GP Practice
Medicine Classification	BNF Chapters

Key Columns Used

Column	Purpose
YEAR_MONTH	Prescribing month
ICB_NAME	NHS Integrated Care Board
PRACTICE_NAME	GP practice
BNF_CHEMICAL_SUBSTANCE	Drug name
BNF_CHAPTER_PLUS_CODE	Drug category classification
ITEMS	Number of prescription items
TOTAL_QUANTITY	Quantity prescribed
ACTUAL_COST	Prescribing expenditure

Methodology

Data Cleaning

The following preprocessing steps were completed:

- Standardised column names
- Checked missing values
- Converted dates into datetime format
- Converted numeric prescribing variables
- Removed invalid or unusable records
- Filled missing address values with "Unknown"

Mental Health Medicine Filtering

Mental health medicines were filtered using BNF classification codes:

BNF Code	Medicine Group
0401	Anxiolytics
0403	Antidepressants

A new categorical variable called Drug_Category was created.

Analytical Components

The project included:

- Exploratory Data Analysis (EDA)
- Monthly prescribing trend analysis
- Yearly prescribing trend analysis
- ICB-level comparison
- Practice-level comparison
- Cost trend analysis
- Top prescribed medicine analysis
- Forecasting using moving averages
- Dashboard dataset generation

Exploratory Data Analysis (EDA)

Dataset Structure

The sample dataset contained:

Metric	Value
Rows analysed (sample)	1,000
Total columns	27
Main data types	Object + Numeric

The dataset included prescribing, organisation, geographic, and medicine classification variables.

Missing Values

Missing values were relatively limited and mainly associated with address-related fields such as:

- ADDRESS_2
- ADDRESS_4

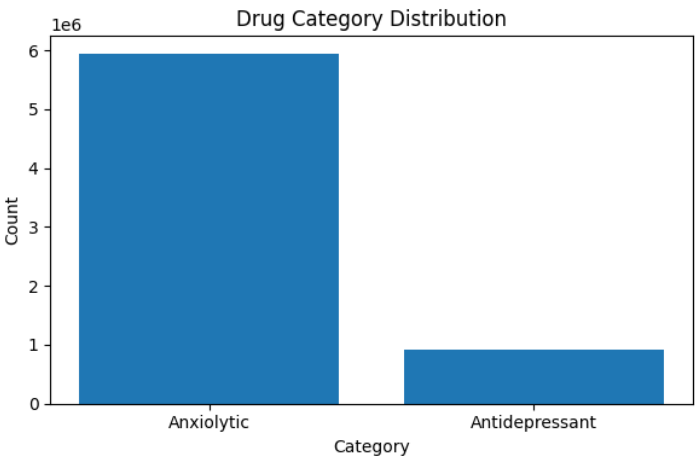
These fields were not critical for prescribing analysis and were filled appropriately.

Drug Category Distribution

The filtered mental health dataset contained two major medicine categories:

- Antidepressants
- Anxiolytics

Antidepressant records were more prevalent than anxiolytic records, indicating higher prescribing activity for depression-related treatment.

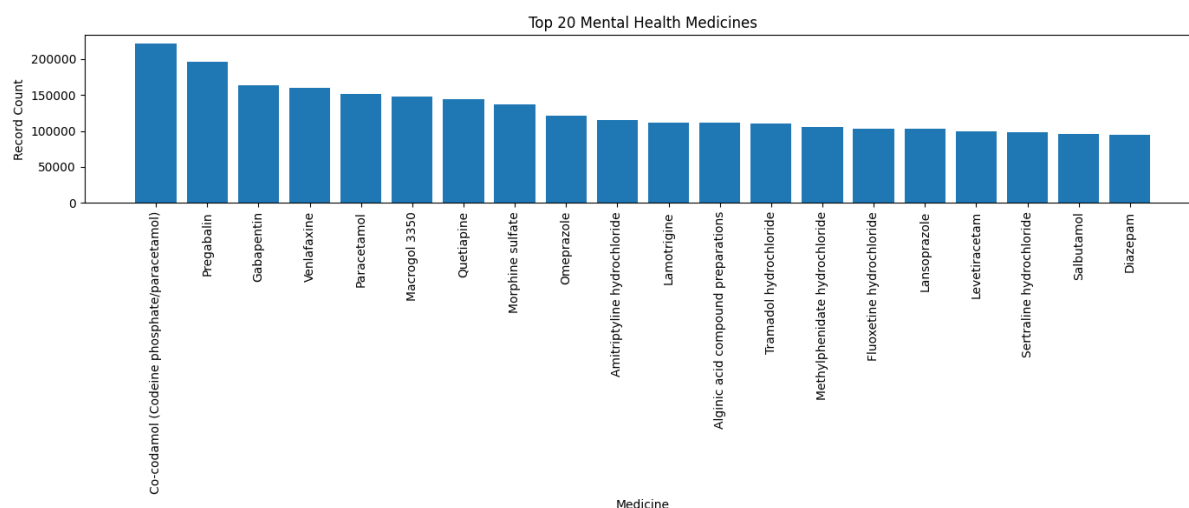


Top Medicines

The most frequently prescribed medicines included commonly used antidepressants and anxiolytics such as:

- Sertraline
- Fluoxetine
- Citalopram
- Diazepam
- Mirtazapine

These medicines accounted for a large proportion of total prescribing activity.



Prescribing Trend Analysis

Monthly Prescribing Trends

Monthly prescribing trends were analysed using:

- total prescription items,
- prescribing quantity,
- and prescribing cost.

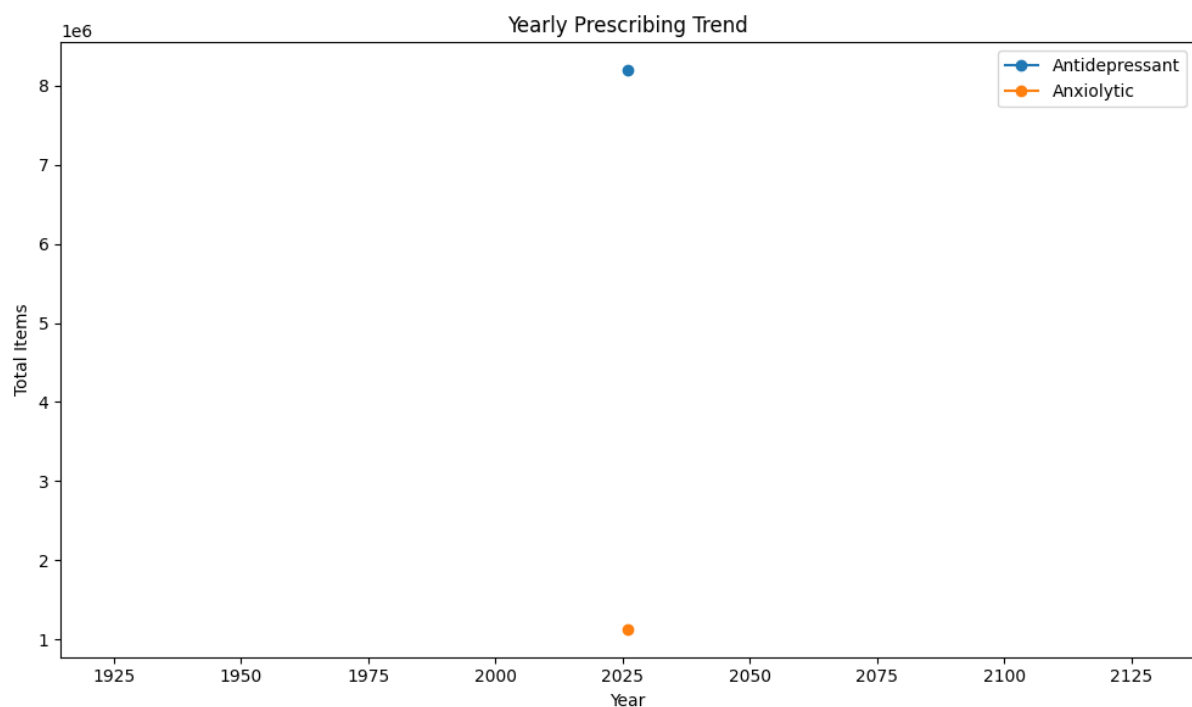
The analysis showed relatively stable prescribing activity over time, with some monthly variation.

Yearly Trends

Yearly aggregation demonstrated long-term prescribing behaviour.

Observations

- Antidepressant prescribing remained the dominant mental health prescribing category.
- Total prescribing items and prescribing cost increased gradually over time.
- Anxiolytic prescribing remained comparatively lower.



ICB-Level Analysis

Prescribing activity was aggregated at Integrated Care Board (ICB) level.

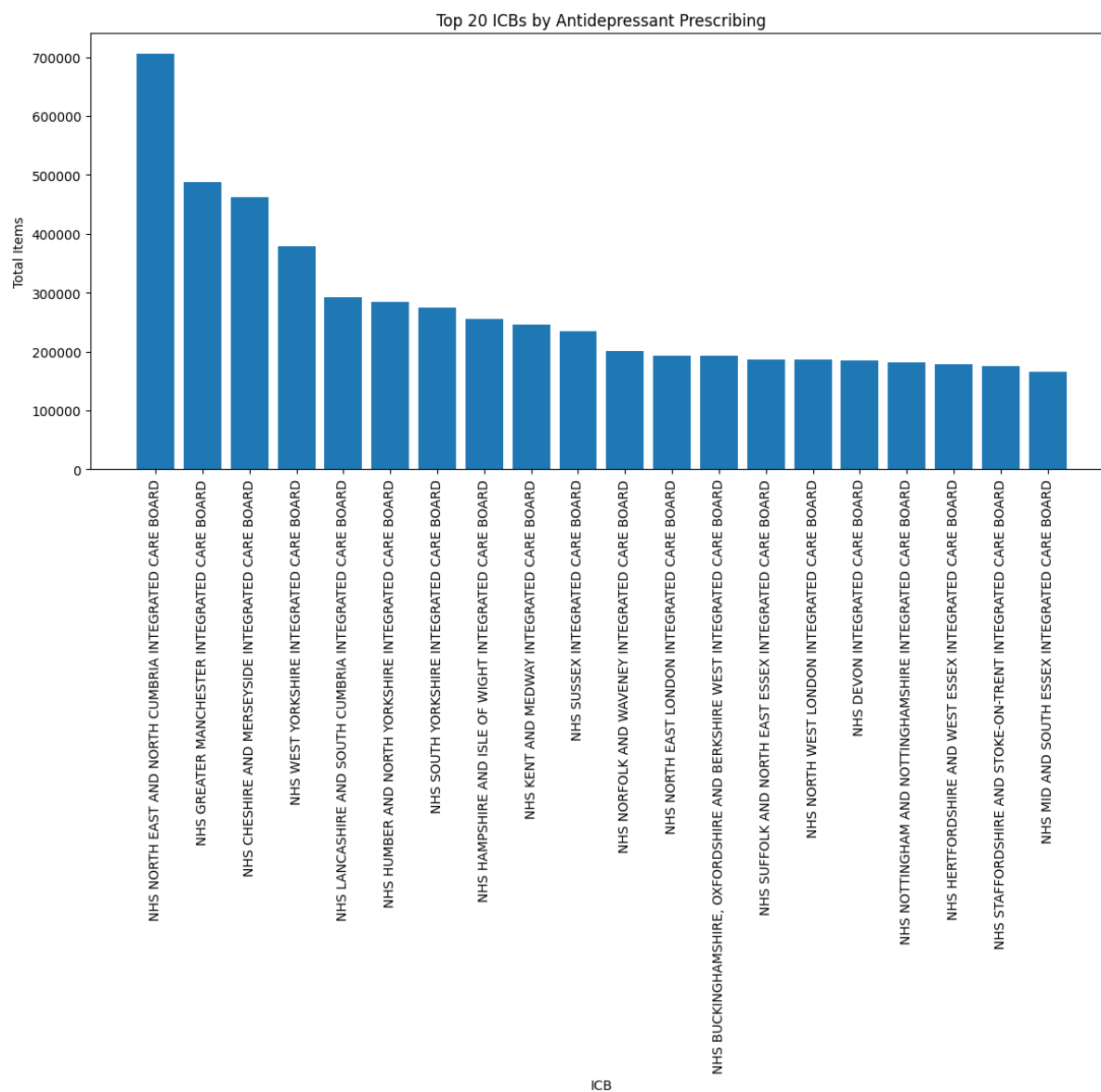
Findings

- Certain ICBs demonstrated substantially higher prescribing volumes.
- Large population regions recorded higher total prescription items and costs.
- Geographic variability suggests differences in:
 - prescribing demand,
 - population size,

- healthcare access,
- and mental health burden.

Top ICBs were visualised using bar charts for:

- antidepressant prescribing,
- anxiolytic prescribing,
- and prescribing cost.

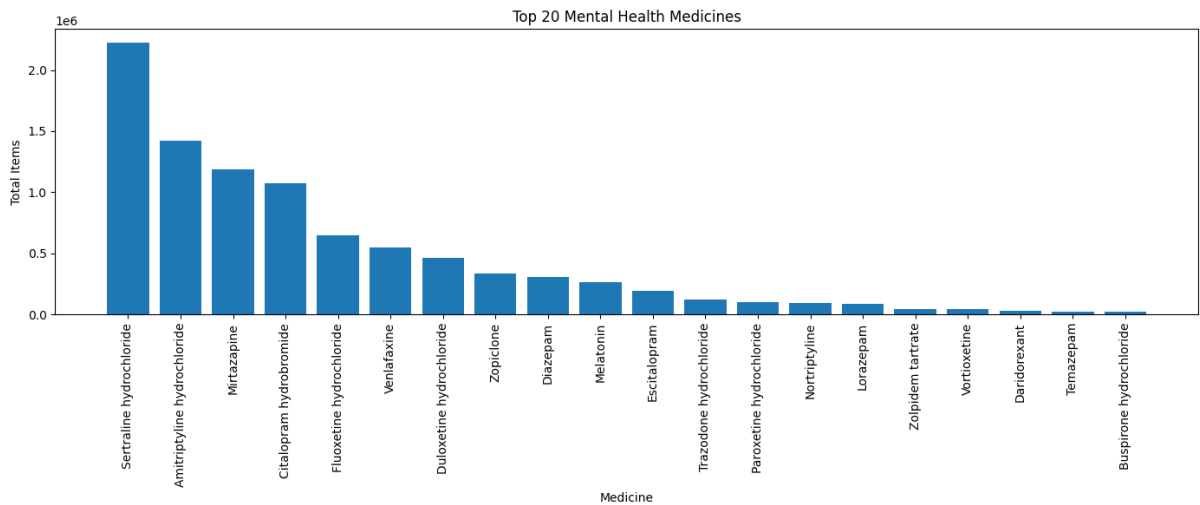


Medicine-Level Analysis

Medicine-level summaries identified the most prescribed mental health medicines.

Key Insights

- SSRI antidepressants accounted for a major share of prescriptions.
- Generic prescribing dominated over branded prescribing.
- Certain medicines generated disproportionately high prescribing costs.



Dashboard Preparation

Dashboard-ready datasets were generated for Power BI/Tableau integration.

Dashboard Files Produced

File	Purpose
dashboard_monthly_trend.csv	Monthly trends
dashboard_icb_summary.csv	ICB comparison
dashboard_drug_summary.csv	Medicine-level analysis

These files support creation of:

- KPI dashboards,
- trend dashboards,
- geographic comparison dashboards,
- and forecasting dashboards.

Visualisations Produced

The project generated multiple visual outputs including:

Visualisation	Purpose
Monthly trend line charts	Trend analysis
Cost trend charts	Expenditure analysis
Top ICB bar charts	Regional comparison
Top medicines charts	Medicine analysis
Heatmaps	Cost comparison
Forecast charts	Demand prediction

Limitations

The analysis has several limitations:

1. Forecasting used a simple moving-average approach rather than advanced time-series modelling.
2. The analysis focused only on prescribing records and did not include clinical outcomes.
3. Population-adjusted prescribing rates were not included.
4. Regional demographic differences were not explicitly modelled.

Conclusion

This project successfully analysed antidepressant and anxiolytic prescribing trends using NHS Open Prescribing data.

Key findings include:

- antidepressants were prescribed more frequently than anxiolytics,
- prescribing activity varied across ICBs,
- prescribing costs remained substantial,
- and mental health medicine demand showed continued stability over time.

The analysis demonstrates how NHS prescribing datasets can support:

- mental health service monitoring,
- prescribing optimisation,
- expenditure tracking,

- and healthcare planning.

The forecasting and dashboard outputs provide additional value for operational reporting and future decision-making.